

Evidential Dirichlet Fusion for Trusted Multimodal Spectrum Sensing

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Abstract—Reliable spectrum sensing under low signal-to-noise ratio (SNR) and unseen modulation conditions remains challenging for cognitive radio and dynamic spectrum access systems. Conventional detectors depend on fixed statistical assumptions or prior signal knowledge, while many deep learning-based methods rely on a single representation and may become overconfident under distribution shift. This paper proposes an evidential multimodal spectrum sensing framework for robust (H_0/H_1) detection. The framework constructs complementary temporal and signal-image representations from IQ samples, extracts sequential and multi-scale spatial features through dual branches, and performs gated cross-attention to calibrate multimodal information before decision making. Instead of using a deterministic softmax classifier, the fused representation is mapped to Dirichlet evidence, providing both sensing probability and epistemic uncertainty. Experiments are conducted on RadioML 2018.01A under a three-stage protocol, including in-distribution validation, unseen-SNR generalization, and unseen-modulation generalization. The proposed model achieves the best ACC, F1-score, and AUC across all three stages, with ACC/F1/AUC values of 0.9507/0.9490/0.9805 on validation, 0.8530/ 0.8316/ 0.8752 on unseen-SNR testing, and 0.8890/ 0.8779/ 0.9166 on unseen-modulation testing. These results show that the proposed framework improves sensing reliability without relying on aggressive (H_1) prediction, especially under low-SNR and blind modulation scenarios.

Keywords—Spectrum sensing, cognitive radio, multimodal learning, evidential deep learning, uncertainty estimation.

I. INTRODUCTION

Spectrum scarcity remains a persistent challenge in modern wireless systems. With the development of cognitive radio, dynamic spectrum access, Internet of Things, and next-generation communication networks, spectrum resources must be used more efficiently rather than being statically allocated. Spectrum sensing is therefore a fundamental task in intelligent wireless systems, where a detector determines whether a spectrum segment is idle or occupied so that secondary users can access available bands without interfering with licensed users.

In binary spectrum sensing, the received signal is usually formulated under two hypotheses: (H_0), where the channel is idle and the observation contains only noise, and (H_1), where the channel is occupied and the observation contains both signal and noise. A reliable detector should identify weak signal presence while maintaining a low false-alarm probability. This requirement becomes difficult in practical

environments, where received signals are affected by propagation loss, fading, interference, hardware impairments, and modulation diversity. Therefore, spectrum sensing is not only an in-distribution classification problem, but also a robustness problem under SNR shift and signal-distribution shift.

Conventional sensing methods, including energy detection, matched filtering, and cyclostationary feature detection, have provided the foundation for early cognitive radio systems [1]–[4]. However, these methods usually rely on fixed statistical assumptions or prior signal knowledge. Energy detection is sensitive to noise uncertainty and threshold mismatch, matched filtering requires accurate waveform knowledge and synchronization, and cyclostationary detection often incurs higher computational complexity and longer observation windows. Their reliability can degrade when the sensing environment violates these assumptions.

Learning-based spectrum sensing has been introduced to reduce dependence on hand-crafted decision rules. Traditional machine learning methods use engineered features to train (H_0/H_1) classifiers, while deep learning methods learn representations from IQ samples or signal-derived images [5]–[7]. Although these models can better adapt to complex channel conditions, many of them still rely on a single signal view. Time-domain models can capture phase and frequency evolution but may be fragile under severe noise corruption. Image-domain models can extract spectral or time-frequency structures but may be sensitive to noise-like textures and lose fine temporal dynamics. This makes single-view learning insufficient for reliable sensing under low-SNR and unseen-modulation conditions.

Recent multimodal sensing methods combine complementary signal views and have shown improved robustness over single-branch models [8]. Nevertheless, two limitations remain. First, many fusion strategies rely on simple concatenation or deterministic attention without explicitly modeling modality reliability. When one modality becomes unreliable under low SNR, misleading features may be propagated to the final decision. Second, most deep detectors output deterministic softmax probabilities, which can be poorly calibrated under unseen SNRs or modulation formats. Such overconfidence is risky in spectrum sensing because it may either increase false alarms or miss weak primary-user signals.

To address these issues, this paper proposes an evidential multimodal spectrum sensing framework. The framework constructs complementary temporal and signal-image representations from the same IQ sample, extracts sequential and multi-scale spatial features through dual branches, and introduces gated cross-attention for feature-level multimodal calibration. Instead of using a deterministic softmax classifier, the fused representation is mapped to Dirichlet evidence, which provides both the (H_1) probability and epistemic uncertainty. This allows the model to make sensing decisions while reflecting evidence strength, especially for weak, ambiguous, or unseen signal patterns.

The main contributions of this paper are summarized as follows. First, an evidential multimodal framework is proposed for robust (H_0/H_1) spectrum sensing under SNR and modulation shifts. Second, complementary temporal and signal-image representations are constructed to jointly characterize phase-frequency dynamics, spectral structures, state transitions, and recurrence patterns. Third, a gated cross-attention module is developed to learn modality reliability at the feature level. Finally, Dirichlet evidential modeling is introduced to improve decision reliability and reduce overconfidence under low-SNR and unseen-modulation conditions.

II. RELATED WORK

A. Conventional Spectrum Sensing

Early cognitive radio studies established spectrum sensing as the key mechanism for opportunistic spectrum access, where secondary users must reliably detect primary-user activity before accessing licensed bands [1], [2]. Conventional spectrum sensing methods mainly rely on hand-crafted statistical decision rules. Energy detection (ED) estimates channel occupancy from received signal energy and does not require prior knowledge of the primary signal. This simplicity makes ED attractive for practical implementation; however, because noise and signal energy become difficult to separate at low SNR, ED is highly sensitive to noise uncertainty and threshold mismatch [3]. Therefore, although ED is computationally efficient, it tends to produce unreliable decisions in precisely the low-SNR regime where robust spectrum sensing is most needed.

Matched filtering (MF) improves detection by correlating the received waveform with a known signal template, and it is statistically efficient when accurate prior information is available [3]. However, this advantage depends on synchronization, waveform knowledge, and channel assumptions that are rarely guaranteed in open wireless environments. In contrast, cyclostationary feature detection (CSD) exploits periodic statistics induced by modulation, coding, pilots, or symbol timing. By detecting cyclic spectral structures, CSD can distinguish modulated signals from stationary noise and is therefore more robust than ED under certain low-SNR conditions [4]. Nevertheless, CSD usually requires longer observation windows, higher computational cost, and careful feature selection. These

conventional methods reveal a common limitation: their confidence is tied to fixed assumptions about noise, signal templates, or cyclic features. Once these assumptions are violated, the detector can become confidently wrong.

B. Learning-Based Spectrum Sensing

Machine learning-based spectrum sensing addresses part of this limitation by replacing manually designed decision statistics with data-driven feature learning. Classical learning models use engineered features such as energy statistics, covariance features, or spectral descriptors to train classifiers for H_0/H_1 decision making. Compared with fixed-threshold detectors, these methods can adapt better to complex channel and noise conditions, but their performance still depends heavily on the quality of manually selected features.

Deep learning further reduces this dependence by learning representations directly from IQ samples or signal-derived images. Lee et al. proposed a deep cooperative sensing framework in which CNNs learn the fusion of sensing results across multiple secondary users, allowing spatial and spectral correlation to be modeled adaptively rather than through explicit mathematical rules [5]. Tekbiyik et al. introduced a CNN-based method using the spectral correlation function, showing that cyclostationary information can be converted into a learnable representation for joint spectrum sensing and signal identification [6]. RNN and BiLSTM-based methods model temporal dependencies in received signals and are particularly useful when the waveform contains sequential structures under fading or low-SNR conditions [7]. These works demonstrate that deep models can substantially improve sensing performance over classical detectors.

However, most learning-based detectors still produce deterministic softmax probabilities. Such outputs may be accurate on validation data but poorly calibrated under distribution shift. In spectrum sensing, this is not a minor issue: a detector that is overconfident under unseen SNRs or unseen modulations can either waste spectrum opportunities by excessive false alarms or miss weak primary-user signals. In addition, single-view learning methods are inherently limited. Time-domain models can capture phase and frequency evolution, but they may be fragile when the waveform is buried in noise. Image-domain models can capture spectral or time-frequency structures, but they may become overly sensitive to noise-like textures. These limitations motivate multimodal and uncertainty-aware spectrum sensing.

C. Multimodal and Uncertainty-Aware Spectrum Sensing

Multimodal spectrum sensing attempts to combine complementary signal views. Recent cross-attention models such as BCAM-Net use bidirectional interactions between temporal and time-frequency branches to improve robustness under low SNR and non-Gaussian noise [8]. This line of work shows that multimodal interaction is more effective than simply relying on a single branch.

Nevertheless, most existing multimodal detectors still perform deterministic fusion: they learn better features, but they do not explicitly indicate how much evidence supports the final decision. As a result, the model may still appear confident even when the input lies outside the training distribution.

Uncertainty-aware deep learning provides a way to address this problem. Evidential deep learning (EDL) models class probabilities as a Dirichlet distribution, allowing the network to output not only an expected class probability but also an evidence strength and epistemic uncertainty [9]. This mechanism is particularly suitable for spectrum sensing because low-SNR and unseen-modulation samples often contain weak or conflicting cues. Instead of forcing the network to make an overconfident decision, Dirichlet evidence modeling allows uncertain samples to carry lower evidence.

The proposed framework follows this direction but differs from existing works in two aspects. First, it performs feature-level multimodal calibration through gated cross-attention, allowing temporal and image features to selectively refine each other before the final decision. Second, it maps the fused representation to Dirichlet evidence, so that the final H_0/H_1 output is accompanied by an uncertainty estimate. Unlike rule-based evidence combination, the proposed method does not require manually designed modality-specific mass functions. Instead, modality reliability and decision evidence are learned jointly from data. This makes the framework more suitable for high-dimensional multimodal spectrum sensing under SNR and modulation shifts.

III. FRAMEWORK OVERVIEW

A. Algorithm Overview

The proposed framework is designed for trusted binary spectrum sensing under SNR shift and modulation shift. Given a complex baseband IQ sequence $\mathbf{x} \in \mathbb{R}^{1024 \times 2}$, the task is to decide whether the observed spectrum segment corresponds to signal absence H_0 or signal presence H_1 . Instead of relying on a single representation, the proposed model constructs two complementary views from the same IQ sample: a temporal representation that preserves phase-frequency evolution, and an image representation that encodes spectral, time-frequency, recurrence, and transition structures.

As illustrated in Fig. 1, the framework contains four stages. First, the raw IQ sample is transformed into a six-channel temporal feature sequence and an eight-channel signal-image tensor. Second, the temporal branch extracts sequential dependencies through channel attention, multi-scale one-dimensional convolution, and temporal convolutional network (TCN) blocks. Third, the image branch learns multi-scale spatial structures through modality attention and a ResNet18-style pyramid backbone. Finally, the two branch embeddings are mutually calibrated by gated

cross-attention and sent to a Dirichlet evidence head, which produces both class probability and epistemic uncertainty.

This design follows a simple principle: weak signal detection should not be decided by a single unstable view. At low SNR, time-domain waveforms may be strongly corrupted by noise, while image-domain structures may still retain weak but useful spectral regularities. Conversely, some modulation families exhibit phase or instantaneous-frequency patterns that are more visible in the temporal stream than in the image stream. Therefore, the model is not built as a simple concatenation network; it is built as a selective multimodal evidence network.

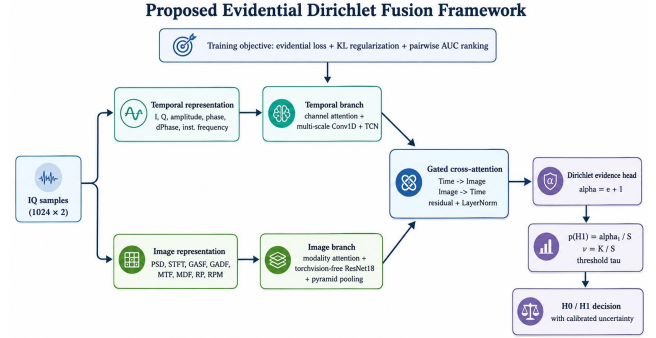


Fig. 1. Overall architecture of the proposed evidential multimodal spectrum sensing framework. The raw IQ sequence is converted into temporal and image representations, fused by gated cross-attention, and mapped to Dirichlet evidence for threshold-based H_0/H_1 sensing.

B. Data Preprocessing and Multimodal Representation

Let the received complex baseband signal be

$$\mathbf{r}[n] = I[n] + jQ[n], \quad n = 1, 2, \dots, N$$

where $N = 1024$, and $I[n]$ and $Q[n]$ denote the in-phase and quadrature components, respectively. The proposed preprocessing stage converts the raw IQ sequence into a temporal feature sequence and an image tensor before network inference.

1) Temporal Representation

The temporal branch uses six physically meaningful channels:

$$\mathbf{x}_t[n] = [I[n], Q[n], A[n], \phi[n], \Delta\phi[n], f_{\text{inst}}[n]]$$

where the amplitude is

$$A[n] = \sqrt{I[n]^2 + Q[n]^2}$$

and the unwrapped phase is

$$\phi[n] = \text{unwrap}(\arctan 2(Q[n], I[n]))$$

The phase difference and instantaneous frequency are computed as

$$\Delta\phi[n] = \phi[n] - \phi[n-1]$$

$$f_{\text{inst}}[n] = \frac{1}{2\pi} \Delta\phi[n]$$

In implementation, f_{inst} is clipped to a bounded range to prevent unstable spikes from dominating the temporal stream. The six-channel feature is then normalized channel-wise. Compared with raw IQ alone, this representation makes phase rotation, frequency drift, and modulation-dependent temporal variation more explicit, which is especially useful for PSK, QAM, APSK, and analog modulation families.

2) Image Representation

The image branch uses eight signal-image modalities:

$$\mathbf{X}_i = [\text{PSD}, \text{STFT}, \text{GASF}, \text{GADF}, \text{MTF}, \text{MDF}, \text{RP}, \text{RPM}]$$

The PSD channel is computed by Welch estimation. Given the complex sequence $r[n]$, the power spectral density can be written as

$$P_r(f) = \mathbb{E}[|R(f)|^2]$$

where $R(f)$ denotes the Fourier-domain representation of $r[n]$. In the implementation, the PSD magnitude is log-scaled and converted into a two-dimensional map through an outer-product construction after normalization.

The STFT channel captures local time-frequency variation:

$$X(\tau, \omega) = \sum_n r[n] w[n - \tau] e^{-j\omega n}$$

where $w[\cdot]$ is the analysis window. The log-amplitude spectrogram is then normalized and resized.

The GASF and GADF channels follow the Gramian angular field idea, which maps a normalized one-dimensional sequence into angular space and models pairwise temporal correlations^[14]. For a normalized amplitude sequence $\tilde{A}[n]$, define

$$\theta_n = \arccos(\tilde{A}[n])$$

The summation and difference fields can be expressed as

$$\text{GASF}_{m,n} = \cos(\theta_m + \theta_n)$$

$$\text{GADF}_{m,n} = \sin(\theta_m - \theta_n)$$

The MTF and MDF channels describe transition behavior of discretized amplitude states. If q_n is the quantized state of the amplitude at time n , the Markov transition probability is

$$M_{a,b} = P(q_{n+1} = b \mid q_n = a)$$

The MTF encodes this transition structure over temporal positions, while the MDF normalizes row-wise transition density to emphasize relative transition patterns.

The RP and RPM channels are based on recurrence analysis^[15]. For a downsampled amplitude sequence, the recurrence plot is defined as

$$\text{RP}_{m,n} = \mathbb{I}(|A[m] - A[n]| \leq \varepsilon)$$

where ε is a recurrence threshold. The RPM channel further emphasizes local recurrence patterns. All eight modalities are resized to a unified spatial size and stacked into an eight-

channel tensor. This design allows the image branch to observe not only spectral energy but also temporal correlation, state transition, and recurrence structures.

C. Model Architecture

The proposed network consists of three core modules: a temporal feature extraction module, an image feature extraction module, and a gated multimodal fusion module. The final output is produced by a Dirichlet evidence head rather than an ordinary softmax classifier.

1) Temporal Feature Extraction Module

The temporal branch is designed to capture both local waveform patterns and longer-range sequential dependencies. Given the temporal input $\mathbf{X}_t \in \mathbb{R}^{T \times 6}$, channel attention is first applied to reweight the six physical channels:

$$\mathbf{w}_t = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \text{AvgPool}(\mathbf{X}_t)))$$

$$\mathbf{X}'_t = \mathbf{X}_t \odot \mathbf{w}_t$$

where $\sigma(\cdot)$ is the sigmoid function, $\delta(\cdot)$ denotes ReLU activation, and $\odot$ denotes channel-wise multiplication. This operation follows the idea of channel recalibration in squeeze-and-excitation style attention^[10].

After channel reweighting, the sequence is processed by a multi-scale one-dimensional convolution front-end. Four parallel convolutional kernels are used:

$$\mathbf{Z}_t = \text{Concat}[\text{Conv}_1(\mathbf{X}'_t), \text{Conv}_3(\mathbf{X}'_t), \text{Conv}_5(\mathbf{X}'_t), \text{Conv}_7(\mathbf{X}'_t)]$$

where the kernel sizes 1, 3, 5, and 7 provide different temporal receptive fields. This design is motivated by the fact that symbol-rate structures, phase transitions, and short-term distortions may appear at different temporal scales.

The multi-scale features are then sent into two TCN residual blocks. A dilated causal convolution can be written as

$$(\mathbf{x} *_{\text{d}} \mathbf{k})[t] = \sum_{m=0}^{K-1} \mathbf{k}[m] \mathbf{x}[t - dm]$$

where d is the dilation factor. The proposed branch uses dilation factors 2 and 4, allowing the model to capture longer temporal dependencies without recurrent computation. A residual TCN block is formulated as

$$\mathbf{H}_{l+1} = \mathbf{H}_l + \mathcal{F}_{\text{TCN}}(\mathbf{H}_l)$$

where $\mathcal{F}_{\text{TCN}}$ contains dilated convolution, batch normalization, GELU activation, and dropout. Finally, global average pooling over the temporal dimension produces the temporal embedding

$$\mathbf{f}_t \in \mathbb{R}^d$$

In the implementation, $d = 160$. The use of TCN is motivated by its ability to model long sequential dependencies with stable parallel convolutional computation^[11].

2) Image Feature Extraction Module

The image branch receives the eight-channel tensor $X_i \in \mathbb{R}^{8 \times H \times W}$. Before spatial feature extraction, a two-dimensional channel attention module reweights the eight modalities. Unlike the temporal attention, this module uses both average pooling and max pooling:

$$\mathbf{w}_i = \sigma(\text{MLP}(\text{AvgPool}(\mathbf{X}_i)) + \text{MLP}(\text{MaxPool}(\mathbf{X}_i)))$$

$$\mathbf{X}'_i = \mathbf{X}_i \odot \mathbf{w}_i$$

This allows the network to adaptively emphasize more reliable image modalities. For example, PSD and STFT may be informative when spectral energy is visible, while recurrence and transition maps may preserve structure when direct spectral contrast is weak.

The reweighted tensor is passed into a ResNet18-style backbone implemented purely in PyTorch. The backbone contains a 7×7 initial convolution, max-pooling, and four residual stages. Each residual block follows

$$\mathbf{Y} = \delta(\mathbf{X} + \mathcal{G}(\mathbf{X}))$$

where $\mathcal{G}(\cdot)$ denotes two convolution-normalization layers, and $\delta(\cdot)$ denotes ReLU activation. Residual learning stabilizes deep feature extraction and alleviates degradation when stacking convolutional layers [12].

Instead of using only the final ResNet layer, the implementation extracts multi-scale feature maps from layer2, layer3, and layer4:

$$\mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4 = \text{Backbone}(\mathbf{X}'_i)$$

Each feature map is aggregated by adaptive average pooling:

$$\mathbf{p}_l = \text{GAP}(\mathbf{F}_l), \quad l \in \{2, 3, 4\}$$

The pyramid representation is then formed as

$$\mathbf{p}_{\text{img}} = [\mathbf{p}_2; \mathbf{p}_3; \mathbf{p}_4]$$

where $[\cdot; \cdot]$ denotes concatenation. In the code, the three pooled vectors have dimensions 128, 256, and 512, resulting in an 896-dimensional pyramid feature. A projection layer maps it into the shared embedding space:

$$\mathbf{f}_i = \text{Proj}(\mathbf{p}_{\text{img}}) \in \mathbb{R}^d$$

This pyramid design is important because weak-signal structures are not confined to a single spatial scale. Low-level time-frequency textures, mid-level spectral transitions, and high-level global patterns may all contribute to the final H_0/H_1 decision.

3) Gated Multimodal Fusion Module

After obtaining the temporal embedding $\mathbf{f}_t$ and image embedding $\mathbf{f}_i$, the model performs bidirectional cross-modal calibration through gated cross-attention. The scaled dot-product attention is defined as

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}$$

The temporal-to-image context is

$$\mathbf{c}_{t \leftarrow i} = \text{Attn}(\mathbf{W}_q^t \mathbf{f}_t, \mathbf{W}_k^i \mathbf{f}_i, \mathbf{W}_v^i \mathbf{f}_i)$$

and the image-to-temporal context is

$$\mathbf{c}_{i \leftarrow t} = \text{Attn}(\mathbf{W}_q^i \mathbf{f}_i, \mathbf{W}_k^t \mathbf{f}_t, \mathbf{W}_v^t \mathbf{f}_t)$$

Two learnable gates control how much cross-modal context is injected:

$$\tilde{\mathbf{f}}_t = \text{LN}(\mathbf{f}_t + \sigma(g_t) \mathbf{c}_{t \leftarrow i})$$

$$\tilde{\mathbf{f}}_i = \text{LN}(\mathbf{f}_i + \sigma(g_i) \mathbf{c}_{i \leftarrow t})$$

where g_t and g_i are learnable scalar gates, and $\text{LN}(\cdot)$ denotes layer normalization. The fused representation is computed as

$$\mathbf{f}_{\text{fuse}} = \frac{1}{2}(\tilde{\mathbf{f}}_t + \tilde{\mathbf{f}}_i)$$

The final classifier receives the concatenated representation

$$\mathbf{z} = [\mathbf{f}_t; \mathbf{f}_i; \mathbf{f}_{\text{fuse}}]$$

which preserves the original temporal evidence, original image evidence, and cross-calibrated multimodal evidence. This differs from simple feature concatenation because each modality is first refined using information from the other modality, and the influence of cross-modal context is controlled by learnable gates.

4) Dirichlet Evidence Head and Uncertainty Interpretation

Instead of producing unconstrained logits followed by softmax, the model outputs non-negative evidence values. Let

$$\mathbf{e} = [e_0, e_1], \quad e_k \geq 0$$

where e_0 and e_1 correspond to H_0 and H_1 , respectively. In the implementation, evidence is obtained by applying softplus to the final linear output:

$$e_k = \text{softplus}(o_k) + \epsilon$$

The Dirichlet parameter is then defined as

$$\alpha_k = e_k + 1$$

The total evidence strength is

$$S = \sum_{k=1}^K \alpha_k$$

and the expected class probability is

$$p_k = \frac{\alpha_k}{S}$$

The epistemic uncertainty is

$$u = \frac{K}{S}$$

For binary sensing, $K = 2$, and the sensing score is

$$p(H_1 | x) = p_1 = \frac{\alpha_1}{\alpha_0 + \alpha_1}$$

This formulation follows evidential deep learning, where the network learns a distribution over class probabilities rather than only a point estimate [13]. It also admits a Dempster-Shafer / subjective-logic interpretation. The belief mass assigned to class k can be written as

$$b_k = \frac{e_k}{S}$$

while the remaining uncertainty mass is

$$u = \frac{K}{S}$$

Thus,

$$\sum_{k=1}^K b_k + u = 1$$

This interpretation is important for spectrum sensing. When the model has strong and consistent evidence, S becomes large and u becomes small. When the input is ambiguous, such as in extremely low-SNR regions or under unseen modulation formats, the evidence strength is lower and the uncertainty increases. Therefore, the model is not forced to be overconfident on samples whose signal structure is intrinsically weak or unfamiliar.

5) Clarification: Relation to Explicit Dempster-Shafer Fusion

The uncertainty representation above is related to subjective logic and Dempster-Shafer belief assignment [18], [19], but the proposed model does not explicitly perform Dempster-Shafer posterior fusion between the temporal and image branches. In an explicit D-S fusion system, the temporal and image branches would first produce separate mass functions, such as $m_t(H_0)$, $m_t(H_1)$, $m_t(\Theta)$ and $m_i(H_0)$, $m_i(H_1)$, $m_i(\Theta)$. Their conflict coefficient and fused mass would then be computed by rules such as

$$K = \sum_{B \cap C = \emptyset} m_t(B)m_i(C)$$

$$m(H) = \frac{\sum_{B \cap C = H} m_t(B)m_i(C)}{1 - K}$$

Our implementation follows a different route. Temporal and image features are fused at the feature level by gated cross-attention, and the fused representation is then mapped to a single Dirichlet distribution. This is more suitable for the present high-dimensional deep multimodal setting, because it avoids hand-crafted modality-specific mass assignment and allows modality reliability to be learned end-to-end from data.

Table 1. Difference between explicit D-S fusion and the proposed evidential multimodal fusion.

Aspect	Explicit D-S Fusion	Proposed Method
Fusion level	Evidence/posterior level	Feature level + Dirichlet output
Mass assignment	Requires hand-crafted or branch-specific mass functions	Evidence is learned end-to-end
Modality reliability	Controlled by predefined	Learned by gated

	rules	cross-attention
Uncertainty	Explicit mass on ignorance Θ	Epistemic uncertainty $u = K/S$
Suitability	Interpretable but sensitive to rule design	Better suited to high-dimensional deep multimodal sensing

D. Training Strategy

The training objective combines evidential learning, uncertainty regularization, class-aware reweighting, and pairwise ranking. This design is motivated by the evaluation target: the model must not only classify validation samples correctly, but also preserve separability between H_0 and H_1 under SNR shift and modulation shift.

1) Evidential Loss

Given the one-hot label y and Dirichlet expected probability p , the evidential error term is

$$\mathcal{L}_{\text{err}} = \sum_{k=1}^K (y_k - p_k)^2$$

The Dirichlet predictive variance is

$$\mathcal{L}_{\text{var}} = \sum_{k=1}^K \frac{p_k(1-p_k)}{S+1}$$

The basic evidential loss is

$$\mathcal{L}_{\text{EDL}} = \mathcal{L}_{\text{err}} + \mathcal{L}_{\text{var}}$$

The variance term prevents the model from increasing evidence blindly. A wrong but highly confident prediction receives a larger penalty than a wrong but uncertain prediction. This property is particularly relevant for low-SNR spectrum sensing, where overconfident false alarms are harmful.

2) Focal Reweighting for Hard Low-SNR Samples

To focus training on difficult samples, the evidential loss is multiplied by a focal factor:

$$\mathcal{L}_{\text{focal}} = w_y(1 - p_y)^\gamma \mathcal{L}_{\text{EDL}}$$

where p_y is the predicted probability of the ground-truth class, w_y is the class weight, and γ is the focusing parameter. In the implementation, $\gamma = 1.5$, and the class weights are set to (1.0, 1.15), giving slightly higher emphasis to H_1 samples. This is consistent with the goal of improving detection probability without losing control of false alarms.

3) KL Regularization on Unsupported Evidence

To suppress unsupported evidence on non-target classes, a KL regularization term is applied to a modified Dirichlet parameter:

$$\tilde{\alpha} = y + (1 - y) \odot \alpha$$

The KL divergence to a uniform Dirichlet prior $\text{Dir}(1)$ is

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}[\text{Dir}(\tilde{\alpha}) \parallel \text{Dir}(\mathbf{1})]$$

This term discourages the model from assigning excessive evidence to incorrect classes. In the code, the KL coefficient is set to

$$\lambda_{KL} = 0.003$$

4) Pairwise AUC Ranking Loss

Since spectrum sensing is highly sensitive to threshold selection, the training objective further includes a pairwise ranking loss to improve the separability between H_1 and H_0 . The score is defined as the log-odds between the two Dirichlet expected probabilities:

$$s(x) = \log p_1(x) - \log p_0(x)$$

For positive samples $i \in P$ and negative samples $j \in N$, the ranking loss is

$$\mathcal{L}_{\text{rank}} = \frac{1}{|P||N|} \sum_{i \in P} \sum_{j \in N} \max(0, m - (s_i - s_j))$$

where m is the margin. In the implementation, $m = 0.20$ and

$$\lambda_{AUC} = 0.12$$

This loss directly encourages H_1 samples to receive higher scores than H_0 samples, which is aligned with AUC and threshold-based sensing. The total loss is

$$\mathcal{L} = \mathbb{E}[\mathcal{L}_{\text{focal}}] + \lambda_{KL}\mathcal{L}_{KL} + \lambda_{AUC}\mathcal{L}_{\text{rank}}$$

5) Optimization and Stabilization

The model is trained with AdamW, using module-wise learning rates^[16]. The image backbone uses a smaller learning rate, while the temporal branch, modality projection layers, gated cross-attention, and classifier are trained with larger learning rates. This prevents the convolutional image backbone from being destabilized while allowing the newly added multimodal and evidential modules to adapt quickly.

The learning rate schedule consists of warm-up followed by cosine annealing. For epoch t , the learning rate multiplier is

$$\eta(t) = \frac{t+1}{T_w}, \quad t < T_w$$

$$\eta(t) = \frac{1}{2} \left(1 + \cos \left(\pi \frac{t - T_w}{T - T_w} \right) \right), \quad t \geq T_w$$

where T_w is the warm-up length and T is the maximum number of epochs. In the implementation, $T_w = 5$ and $T = 55$.

To improve training stability, gradient clipping is applied:

$$\|\nabla_{\theta} \mathcal{L}\|_2 \leftarrow \min(\|\nabla_{\theta} \mathcal{L}\|_2, 0.5)$$

and an exponential moving average of model parameters is maintained:

$$\theta_{\text{EMA}}^{(t)} = \rho \theta_{\text{EMA}}^{(t-1)} + (1 - \rho) \theta^{(t)}$$

where $\rho = 0.995$. The checkpoint is selected using a validation detection score rather than training loss alone:

$$\text{Score} = 0.25 \text{ ACC} + 0.40 \text{ F1} + 0.35 \text{ AUC}$$

This is consistent with the sensing objective, because a model with lower training loss is not necessarily the best detector under threshold-based evaluation.

6) Validation-Calibrated Decision Threshold

During inference, the final decision is not forced to use a fixed threshold of 0.5. Instead, the threshold is selected on the validation set and then fixed for Test Set 1 and Test Set 2. The decision rule is

$$\hat{y} = 1 \quad \text{if} \quad p(\mathcal{H}_1 | x) \geq \tau$$

$$\hat{y} = 0 \quad \text{if} \quad p(\mathcal{H}_1 | x) < \tau$$

The threshold-search objective is recall-aware but includes a false-alarm guard:

$$J(\tau) = 0.20 \text{ ACC} + 0.28 \text{ F1} + 0.12 \text{ Precision} + 0.28 P_d + 0.12 \text{ AUC} - 0.06 P_{fa} - \lambda_p \max(0, P_{fa} - 0.035)$$

where

$$\lambda_p = 1.2$$

This threshold calibration follows the spirit of Neyman-Pearson detection^[17]: the operating point should improve detection probability while preventing uncontrolled false alarms. Importantly, the threshold is selected only from the validation set, so no test-label information is used during calibration.

IV. PERFORMANCE EVALUATION

A. Dataset and Evaluation Protocol

The experiments were conducted on the RadioML 2018.01A dataset, a standard large-scale modulation dataset generated from GNU Radio simulations. The original dataset contains 2,555,904 complex baseband samples from 24 modulation types. Each sample is stored in HDF5 format as an IQ sequence X ($N, 1024, 2$), with a one-hot modulation label Y ($N, 24$) and an SNR label Z ($N, 1$). The SNR range spans from -20 dB to 30 dB with a step size of 2 dB.

To make the evaluation computationally tractable while preserving SNR and modulation diversity, we performed stratified sampling with 2000 samples per modulation type, resulting in approximately 48,000 selected samples. The evaluation was deliberately organized into three stages rather than a single random split, because a random split mainly measures in-distribution fitting and cannot reveal whether a detector generalizes to unseen SNRs or unseen modulation formats.

The known modulation set used for training and validation includes BPSK, QPSK, 8PSK, 16PSK, OOK, 4ASK, 8ASK, 16QAM, 32QAM, 64QAM, AM-DSB-SC, AM-SSB-SC, GMSK, and OQPSK. The unknown modulation set used only for blind testing includes 32PSK, 16APSK, 32APSK, 64APSK, 128APSK, 128QAM, 256QAM, AM-SSB-WC, AM-DSB-WC, and FM. The H_1 class denotes signal presence, while the H_0 class is constructed from extremely low-SNR samples at -20 dB and additional AWGN-only samples.

The three-stage evaluation protocol is summarized in Table 2. Instead of using a single random split, the evaluation is deliberately separated into in-distribution validation, unseen-SNR testing, and unseen-modulation testing. This design allows us to examine whether a detector merely fits the training distribution or can remain reliable under practical distribution shifts.

Table 2. Three-stage evaluation protocol.

Stage	H ₁ Source	H ₀ Source	Primary Objective
Validation Set	Known modulations within the training SNR range ([-10,10]) dB	Known modulations at (-20) dB and AWGN-only samples	In-distribution sensing performance
Test Set 1	Known modulations with SNR values outside the training range	AWGN-only samples	SNR generalization
Test Set 2	Unknown modulations over the non-(H ₀) SNR range	Unknown modulations at (-20) dB and AWGN-only samples	Modulation generalization / blind sensing

The validation set measures the basic sensing capability under the same modulation and SNR regime as training. Test Set 1 isolates SNR generalization by keeping the modulation families known while shifting the SNR distribution. Test Set 2 further evaluates blind sensing, where the detector must identify signal presence for modulation families that are never observed during training.

B. Baseline Models

CNN baseline. The CNN baseline represents image-domain spectrum sensing. Prior CNN-based spectrum sensing work has shown that convolutional models can learn discriminative spectral or cyclostationary representations without relying on manually designed statistical decision rules. Tekbiyik et al. proposed a CNN model based on the spectral correlation function for spectrum sensing and signal identification, showing that CNN-based sensing can reduce the need for prior signal features and post-processing statistical decisions [20]. In our implementation, the CNN baseline uses only the constructed eight-channel signal-image representation and therefore tests how far image-domain features alone can support H₀/H₁ sensing.

BiLSTM baseline. The BiLSTM baseline represents temporal sequence modeling. RNN-Bi-LSTM spectrum sensing has been used to capture bidirectional temporal dependencies in received signals, particularly under fading and low-SNR channel conditions. Kumar et al. reported that RNN-Bi-LSTM improves detection probability and false-alarm behavior by exploiting both past and future sequential dependencies in NOMA waveform sensing [21]. In our comparison, the BiLSTM baseline uses temporal IQ-derived features only, serving as a strong time-domain detector without image-domain structural information.

BCAM-Net baseline. BCAM-Net is a recent multimodal spectrum sensing model based on bidirectional cross-attention. It uses a temporal branch and a time-frequency branch, and performs mutual calibration between the two modalities to improve robustness under low SNR and non-Gaussian impulsive noise [22]. Since BCAM-Net is structurally close to the idea of cross-modal sensing, it is a particularly important baseline for testing whether evidential Dirichlet fusion improves beyond deterministic bidirectional attention.

C. Experimental Setup

All models were trained and evaluated under the same three-stage protocol. The proposed model follows the architecture described in Section III, including a six-channel temporal branch, an eight-channel image branch, gated cross-attention fusion, and a Dirichlet evidence head. To avoid repetition, this section focuses on the experimental configuration rather than re-deriving the model formulation.

Training used AdamW with module-wise learning rates. The image backbone used a learning rate of 5×10^{-5} , while the newly added image projection, temporal branch, gated cross-attention module, and classifier used learning rates from 7×10^{-4} to 9×10^{-4} . The batch size was 64. The maximum number of training epochs was 55, with 5 warm-up epochs and an early stopping patience of 12. During training, IQ-domain augmentation was applied only to the training set, including random phase rotation, time shift, mild carrier-frequency offset, IQ imbalance, amplitude scaling, and low-SNR AWGN hard-positive augmentation. Validation and test sets remained clean to avoid evaluation leakage.

For the proposed evidential model, the final (H₁) score and epistemic uncertainty follow the Dirichlet evidence formulation defined in Section III-C.4. The validation-calibrated decision threshold follows the rule described in Section III-D.6 and is selected only on the validation set before being fixed for Test Set 1 and Test Set 2.

D. Evaluation Metrics

The sensing task is formulated as binary hypothesis testing. Let TP denote H₁ samples correctly detected as H₁, TN denote H₀ samples correctly detected as H₀, FP denote H₀ samples falsely detected as H₁, and FN denote H₁ samples missed as H₀. We report Accuracy, F1-score, Precision, Recall /detection probability P_d, false-alarm probability P_{fa}, and AUC.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$P_d = Recall = \frac{TP}{TP + FN}, \quad P_{fa} = \frac{FP}{FP + TN}$$

$$F1 = \frac{2 \cdot Precision \cdot P_d}{Precision + P_d}$$

Here, P_d measures the probability of detecting an occupied spectrum, while P_{fa} measures the probability of falsely declaring signal presence under H₀. A good spectrum sensing model should not merely maximize P_d; it must also

maintain a low P_{fa} , because excessive false alarms reduce spectrum utilization.

Table 3. Performance comparison of different models on three evaluation sets.

Model	Metric	Validation	Test Set 1	Test Set 2
Ours	ACC	0.9507	0.8530	0.8890
	F1-score	0.9490	0.8316	0.8779
	Precision	0.9813	0.9732	0.9752
	Recall (Pd)	0.9188	0.7260	0.7983
	Pfa	0.0175	0.0200	0.0203
	AUC	0.9805	0.8752	0.9166
BCAM-Net	ACC	0.8642	0.8206	0.8362
	F1-score	0.8481	0.7896	0.8114
	Precision	0.9616	0.9545	0.9559
	Recall (Pd)	0.7587	0.6733	0.7049
	Pfa	0.0303	0.0321	0.0325
	AUC	0.9256	0.8590	0.8922
CNN	ACC	0.8379	0.7862	0.8075
	F1-score	0.8446	0.7850	0.8106
	Precision	0.8112	0.7895	0.7978
	Recall (Pd)	0.8808	0.7805	0.8238
	Pfa	0.2050	0.2081	0.2088
	AUC	0.9304	0.8615	0.8923
BiLSTM	ACC	0.9231	0.8526	0.8808
	F1-score	0.9180	0.8304	0.8674
	Precision	0.9834	0.9778	0.9776
	Recall (Pd)	0.8607	0.7216	0.7795
	Pfa	0.0145	0.0164	0.0178
	AUC	0.9644	0.8650	0.9074

E. Overall Quantitative Comparison

Table 3 reports the quantitative comparison across the three evaluation sets. The proposed model achieves the best ACC, F1-score, and AUC on all three sets. This is important because these three metrics reflect different aspects of sensing behavior: ACC measures global correctness, F1-score balances missed detections and false alarms, and AUC evaluates threshold-independent ranking quality.

On the validation set, the proposed model reaches an ACC of 0.9507, an F1-score of 0.9490, and an AUC of 0.9805. Compared with BCAM-Net, it improves ACC by 8.65 percentage points, F1-score by 10.09 percentage points, and AUC by 5.49 percentage points. Compared with the CNN baseline, the improvements are 11.28, 10.44, and 5.01 percentage points, respectively. Compared with BiLSTM, the gains are smaller but still consistent: 2.76 percentage points in ACC, 3.10 percentage points in F1-score, and 1.61 percentage points in AUC. These results show that the proposed framework does not simply benefit from a larger representation; rather, it combines temporal dynamics, image-domain structure, and evidential fusion in a more effective way.

The false-alarm behavior is also meaningful. The CNN baseline obtains a relatively high validation recall of 0.8808, but its P_{fa} reaches 0.2050. This indicates an aggressive decision boundary: the model detects many H_1 samples, but also frequently mistakes H_0 for H_1 . In contrast, the proposed method achieves a higher recall of 0.9188 while keeping P_{fa} at 0.0175. Therefore, the performance gain is not obtained by simply over-reporting spectrum occupancy.

F. Unseen-SNR Generalization

Test Set 1 evaluates SNR generalization. The model is trained on known modulations within $[-10, 10]$ dB, but tested on SNR values outside this range. This protocol is more demanding than ordinary validation, because the detector

must extrapolate to signal quality conditions not observed during training.

On Test Set 1, the proposed model achieves the highest ACC of 0.8530, F1-score of 0.8316, and AUC of 0.8752. Compared with BCAM-Net, it improves ACC by 3.24 percentage points, F1-score by 4.20 percentage points, and AUC by 1.62 percentage points. Compared with BiLSTM, the ACC and F1-score gains are modest, but the proposed model still improves AUC by 1.02 percentage points. This indicates that even when final operating-point metrics are close, the evidential fusion model provides a better global ranking between H_0 and H_1 samples.

The CNN baseline has the highest P_d on Test Set 1, reaching 0.7805, but this comes with a P_{fa} of 0.2081. Such a high false-alarm rate is unacceptable for conservative spectrum sensing. The proposed model instead achieves a more balanced operating point: $P_d=0.7260$, $P_{fa}=0.0200$, and the best AUC among all methods. In other words, the proposed model does not maximize detection by sacrificing idle-channel availability. It preserves a low false-alarm regime while maintaining the best overall sensing performance.

This behavior supports the role of train-time low-SNR augmentation and Dirichlet evidence modeling. Under SNR shift, deterministic models can become overconfident because they treat uncertain low-SNR samples as ordinary inputs. The proposed model, by contrast, learns to represent both probability and evidence strength, which improves robustness when the test SNR distribution deviates from the training interval.

G. Low-SNR Transition Behavior

The heatmaps provide a finer view of the low-SNR transition region. Unlike aggregate metrics, which average over all SNRs and modulations, the heatmaps reveal where each detector begins to reliably distinguish H_1 from H_0 . For this reason, the heatmaps are not used as decorative figures; they directly expose the transition boundary of each detector under blind modulation conditions.

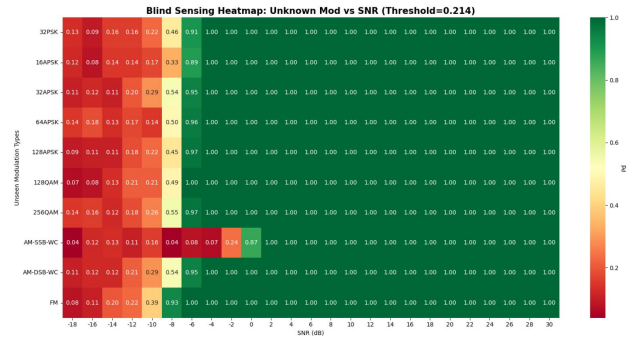


Fig. 2. Blind sensing heatmap of BCAM-Net on unseen modulations. The threshold is 0.214.

For BCAM-Net, most unknown modulations remain weakly detected below -10 dB. Detection improves around -8 dB and becomes stable from approximately -6 dB for most modulation types. However, the transition is not uniform. AM-SSB-WC remains difficult over a wider SNR range,

with low P_d values extending from -8 dB to -4 dB before gradually recovering.

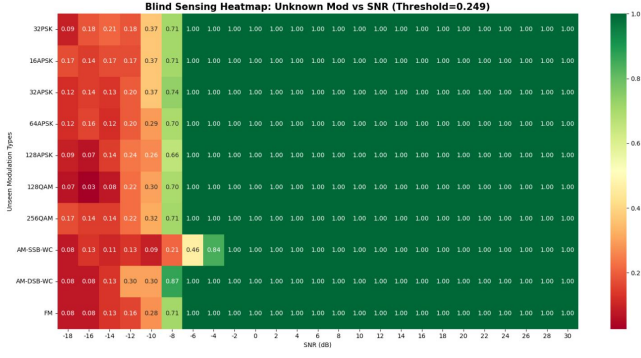


Fig. 3. Blind sensing heatmap of BiLSTM on unseen modulations. The threshold is 0.249.

BiLSTM shows a sharper transition for many digital modulations. Most PSK, APSK, and QAM classes reach reliable detection around -8 dB to -6 dB. Nevertheless, the AM-SSB-WC row remains a persistent failure case. This indicates that a purely temporal detector can learn strong sequential patterns, but its generalization is uneven when the unseen waveform family has different amplitude and spectral structure.

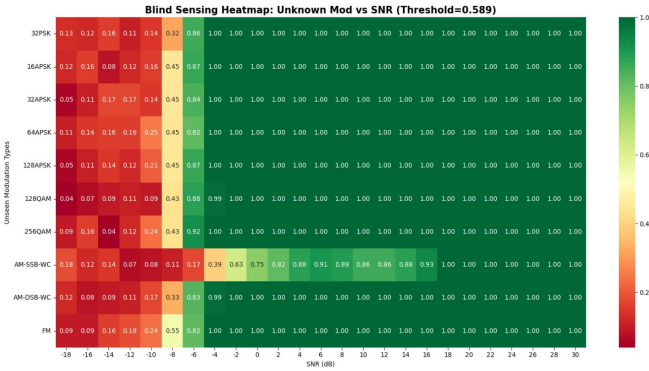


Fig. 4. Blind sensing heatmap of CNN on unseen modulations. The threshold is 0.589.

The CNN heatmap shows a different pattern. It becomes highly confident for most unknown modulations once SNR reaches approximately -6 dB, and it appears visually strong over much of the heatmap. However, Table 1 reveals the hidden cost: CNN has $P_{fa}=0.2088$ on Test Set 2. Therefore, the strong green region in the heatmap must be interpreted together with the false-alarm metric. The CNN detector is sensitive to signal-like image textures, but it also tends to classify noise-only samples as occupied spectrum.

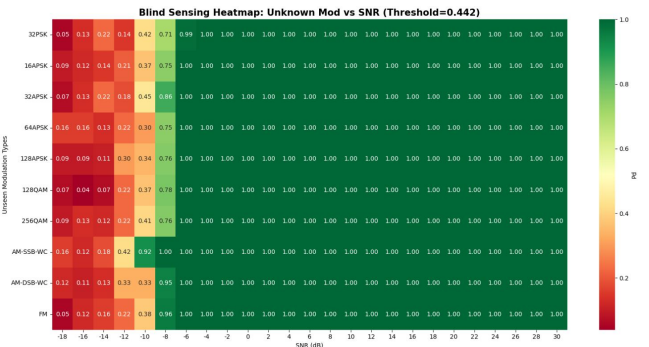


Fig. 5. Blind sensing heatmap of the proposed evidential multimodal model on unseen modulations. The threshold is 0.442.

The proposed model shows the most favorable low-SNR transition. In Fig. 5, most digital modulation families move from weak detection to near-complete detection between -10 dB and -6 dB. At -8 dB, several classes already show strong detection, such as AM-SSB-WC at 0.92, AM-DSB-WC at 0.95, and FM at 0.96. By -6 dB, most rows reach or approach 1.00. More importantly, this transition is achieved while keeping the global P_{fa} at only 0.0203. The result indicates that the proposed model does not merely become more aggressive at low SNR. Instead, it learns a more reliable separation between noise-only inputs and weak signal-bearing inputs.

H. Blind Modulation Generalization

Test Set 2 evaluates blind modulation generalization, where all H_1 modulation types are unseen during training. This is the most important setting in the experiment because the model must detect signal existence without relying on closed-set modulation recognition.

On Test Set 2, the proposed method achieves ACC=0.8890, F1-score = 0.8779, Precision = 0.9752, P_d = 0.7983, P_{fa} = 0.0203, and AUC = 0.9166. It outperforms BCAM-Net by 5.28 percentage points in ACC, 6.65 percentage points in F1-score, 9.34 percentage points in P_d , and 2.44 percentage points in AUC. Compared with BiLSTM, it improves ACC by 0.82 percentage points, F1-score by 1.05 percentage points, P_d by 1.88 percentage points, and AUC by 0.92 percentage points. Compared with CNN, it reduces P_{fa} by 18.85 percentage points while improving ACC, F1-score, Precision, and AUC.

This result shows that the proposed model learns a broader notion of signal presence rather than memorizing known modulation labels. Even when the model cannot identify an unseen waveform as a specific modulation class, it can still determine that the received sample contains structured signal energy rather than pure noise. This distinction is central to blind spectrum sensing. The task is not closed-set modulation classification; it is robust H_0/H_1 detection under waveform uncertainty.

The advantage comes from the complementarity of the two branches. The temporal branch captures phase and instantaneous-frequency dynamics, which are useful for distinguishing modulated signals from AWGN. The image branch captures spectral, recurrence, Markov-transition, and time-frequency structures, which remain informative even when the exact modulation type is unknown. The gated cross-attention module then prevents the final classifier from relying blindly on one modality. This is different from simple feature concatenation, where noisy or distorted features can be passed directly into the decision layer without reliability control.

I. Discussion

The experimental evidence leads to four conclusions. First, the proposed model improves ordinary sensing

performance. On the validation set, it obtains the best ACC, F1-score, and AUC, while maintaining a low false-alarm rate. This demonstrates that the multimodal evidential design is effective even under the in-distribution setting.

Second, the model improves SNR generalization. On Test Set 1, the proposed method remains the best model in ACC, F1-score, and AUC under SNR values outside the training range. The gain over BiLSTM is not large in ACC, but the higher AUC indicates better separability across thresholds, which is important for practical sensing systems where the operating point may change.

Third, the model has a cleaner low-SNR transition. The heatmaps show that most unknown modulation types become reliably detectable around the -8 dB to -6 dB region, while the false-alarm probability remains close to 2%. This behavior is more valuable than simply increasing P_d , because aggressive detectors can obtain high recall by misclassifying H_0 samples as H_1 .

Fourth, the model improves blind modulation sensing. Test Set 2 confirms that the proposed framework is not merely learning a closed set of training modulations. It can detect signal presence for unseen PSK, APSK, QAM, AM, and FM variants, which is closer to the requirement of practical cognitive radio systems.

Overall, the results suggest that robust spectrum sensing should be evaluated beyond ordinary validation accuracy. A deployable detector should be judged by its behavior under SNR shift, extreme low-SNR transition, false-alarm control, and blind modulation generalization. Under this stricter evaluation protocol, the proposed evidential Dirichlet fusion framework provides the strongest and most balanced performance among all compared methods.

V. CONCLUSION

This paper presented an evidential multimodal spectrum sensing framework for reliable H_0/H_1 detection under SNR and modulation generalization. The proposed method transforms each IQ sample into two complementary representations: a six-channel temporal representation describing IQ, amplitude, phase, phase difference, and instantaneous frequency; and an eight-channel image representation containing PSD, STFT, GASF, GADF, MTF, MDF, RP, and RPM. The temporal branch captures phase-frequency dynamics through channel attention, multi-scale Conv1D, and TCN blocks, while the image branch extracts multi-scale structural patterns through modality attention and a ResNet18-style pyramid backbone. These two branches are fused by gated cross-attention and then mapped to a Dirichlet evidence head.

The experimental results demonstrate that the proposed method improves not only ordinary validation performance but also more demanding generalization behavior. On the validation set, Test Set 1, and Test Set 2, the proposed model achieves the best ACC, F1-score, and AUC among all compared methods. More importantly, it maintains a low false-alarm probability while preserving competitive

detection probability. The heatmap analysis further shows that the proposed model produces a cleaner low-SNR transition and stronger blind sensing behavior on unseen modulation types. These results indicate that the improvement does not come from aggressive H_1 prediction, but from more reliable multimodal evidence separation between signal-bearing and noise-only samples.

The key conclusion is that robust spectrum sensing should not be evaluated only by in-distribution accuracy. A deployable detector must remain stable under SNR shift, extreme low-SNR conditions, false-alarm constraints, and unseen modulation families. The proposed framework addresses these requirements by combining gated multimodal feature calibration with Dirichlet evidential uncertainty modeling. Compared with conventional detectors that rely on fixed statistical assumptions and deterministic deep models that may become overconfident, the proposed method provides a more trustworthy sensing decision.

Future work will extend this framework in three directions. First, real over-the-air measurements and hardware-impaired signals will be introduced to further test deployment robustness. Second, explicit rejection or human-in-the-loop mechanisms can be built on top of Dirichlet uncertainty to avoid unsafe decisions under highly ambiguous inputs. Third, recent progress in radio foundation models and RadioLLM suggests a promising direction: spectrum sensing may evolve from task-specific H_0/H_1 detection toward a more general radio intelligence system that jointly understands signal presence, modulation structure, interference type, and spectrum access strategy [24]. The proposed evidential multimodal framework can serve as a reliable sensing front-end for this broader radio-intelligence pipeline.

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